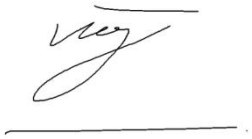
Group Agreement and Plan

A. Group Members

Name	Contact information	Signature
Le Xuan Ny	***Student ID (Australia)***: 105680151 ***Student Email***: 105680151@student.swin.edu.au	
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B. Group style and roles

Role / Position	Team Member	Core Tasks & Responsibilities
Data and Baseline	Hoang	• Collect, inspect, clean, and crop all input facial images. • Apply image transformation techniques (such as rotation, flipping, and brightness adjustment) to enrich the dataset. • Build an automated pipeline to load and process data in batches for optimal speed.

Role / Position	Team Member	Core Tasks & Responsibilities
		• Construct and set up an initial baseline model for user classification tasks. • Train the model, monitor learning progress, and log accuracy evaluation results.
Metric Learning	Duc Anh	• Build a model specialized in extracting distinct feature points from each face. • Establish a mechanism to compare similarities and differences between any two facial images. • Measure feature distance and similarity between images to distinguish individual identities. • Determine optimal decision thresholds to verify if two images belong to the same person. • Plot evaluation curves and summarize measurement metrics for overall system accuracy.
Liveness and Spoof Detection	Pahan	• Prepare datasets simulating genuine users versus fraudulent spoof attempts (printed photos, video replays, or screen displays). • Train models to differentiate real human presence from fake images/videos in real time. • Optimize model processing speeds to ensure smooth performance on live camera feeds. • Establish security verification steps to immediately block spoof attempts early in the pipeline.
Facial Analytics	Hieu	• Collect and organize benchmark image datasets covering various human emotional states. • Resolve sample imbalance issues across different emotion categories. • Design and train a lightweight emotion recognition model optimized for minimal resource usage. • Automatically locate the precise facial region of interest to analyze real-time expressions.
System Integration and GUI	Khai	• Combine individual components into a seamless processing pipeline (Liveness Check ➔ Identity Recognition ➔ Emotion Prediction).

Role / Position	Team Member	Core Tasks & Responsibilities
		• Design the desktop application user interface featuring a live webcam feed. • Display visual bounding boxes around detected faces, identity labels, liveness status, and emotion results. • Manage multi-tasking operations to maintain a smooth camera frame rate without lag.
Project Management and QA	Ny	• Lead the compilation, drafting, and formatting of the complete final project report. • Establish project milestones, monitor team progress, and manage task deadlines. • Inspect and audit code contributions from each team member prior to system integration. • Script, record, edit, and coordinate the final project demonstration video showcase.

C. Timeline

The detailed project timeline and task schedule are managed in Google Sheets.

D. Communication Plan

The detailed communication plan are managed in Google Sheets.

E. On-track behaviours

- Active Participation and Task Ownership:
    - Clear Task Ownership: Each member owns specific modules (Face Verification and Metric Learning, Liveness and Spoof Detection, Emotion Detection, System Integration and UI, Project Management and Video). Responsibilities are tracked publicly on a shared task board.
    - Punctuality and Attendance: Attend all scheduled meetings on time. Members unable to attend must submit written updates beforehand and review meeting notes afterward.
    - Internal Deadlines: Complete all individual deliverables by internal deadlines set several days before official Canvas submissions to allow time for system integration and testing.
    - Version Control and Reproducibility: Commit code regularly using feature branches and pull requests. Log all experimental setups (hyperparameters, metrics like AUC and Accuracy) to ensure reproducible report findings.
    - Communication Window: Respond to group messages within 24 hours, with faster response times during final submission weeks.
    - Individual Record Keeping: Independently log task progress, methodologies, and technical solutions throughout the project to prepare for the individual report and Week 12 interview.
- Proactive Updates and Roadblock Escalation:
    - 30-Minute Weekly Meetings: Share progress using three questions: *What was finished?*, *What is next?*, and *What is blocking me?*
    - Mid-Week Chat Check-Ins: Send brief status updates for assigned tasks (*On Track*, *At Risk*, or *Blocked*).
    - Real-Time Task Tracker: Maintain task statuses (*To Do*, *In Progress*, *In Review*, *Done*) on the shared board (Google Sheets, Notion, or Trello).
    - 24-Hour Early Escalation Rule: Report roadblocks within 24 hours instead of waiting for meetings. Include problem details, attempted fixes, and needed help (e.g., model convergence failures, unseen spoof detection issues, pipeline errors).
    - Module Handover Notes: Document input and output formats, model file paths, and limitations when handing off deliverables to other team members (e.g., embedding outputs to UI).
- Mutual Support and Accountability:
    - Proactive Assistance: Early finishers help teammates with code reviews, validation runs, or physical UI testing using printed photo and screen display samples.
    - Peer Reviews: Require peer review by at least one other member for all core code modules and report sections before final merging or submission.
    - Fair Workload Allocation: Distribute work equitably across all members, ensuring equal speaking time and contribution in the final video demo. Rebalance workloads if someone is overloaded.
    - Handling Slippage: Inform the team in advance with a revised plan if a deadline will be missed. Tasks slipping twice are reassigned or shared, and escalated to the tutor if unresolved.
    - Constructive Feedback: Keep feedback focused strictly on work. Settle technical design choices (e.g., Cosine vs. Euclidean distance) using empirical experimental evidence or a group vote.
    - Integration Sign-Offs: Require full team review and sign-off for each major checkpoint (model training complete, UI integration done, report draft finalized, video rehearsed).
    - Transparent Peer Evaluation: Group marks are adjusted based on peer evaluations; all contributions are tracked transparently throughout the project.

F. Off-track behaviours

The following behaviours may prevent our group from successfully completing the project:

- Missing agreed deadlines or meetings without prior notice, especially when others depend on the delayed work for model comparison, system integration, or report writing.
- Repeatedly failing to respond within our agreed communication timeframe, or not reporting technical difficulties early.
- Contributing incomplete or untested code, failing to share model checkpoints and setup instructions, or making changes that break other modules without informing the group.
- Using fully pretrained models without further training, neglecting either required face verification approach, or leaving the anti-spoofing and emotion detection modules unintegrated.
- Reporting performance without reproducible experiment records, using evaluation data for training, or failing to evaluate face verification using the specified verification pairs and ROC/AUC.
- Leaving integration, testing, documentation, or video preparation until the last minute, creating additional work for other members.
- Contributing unfairly, dismissing constructive feedback, or being unable to explain the methods used in one's assigned work.

When these behaviours occur, we will first contact the member privately to understand the cause and offer support. Together, we will agree on a recovery plan with specific deliverables, a realistic revised deadline, and a follow-up check-in. Support may include debugging together, clarifying requirements, or dividing a task into smaller steps.

We will record the agreed actions in the task tracker and review progress at the next check-in. If missed commitments continue, we will discuss a fair redistribution of urgent tasks and document actual contributions. Repeated or unresolved issues will follow the escalation process below, with the aim of helping the member re-engage while protecting project progress and fairness.

G. Resolving tensions

We will address disagreements early and respectfully, focusing on the issue rather than blaming individuals. Each member will have an opportunity to explain their perspective without interruption. Discussions will refer to the project requirements, agreed responsibilities, and available evidence. In terms of technical disagreements, such as model selection, similarity metrics, or module integration, we will compare options through small experiments using consistent evaluation conditions. Decisions will consider performance, implementation effort, compatibility, and the remaining time. For face verification, ROC/AUC results will inform our choice alongside the practical needs of the attendance system. Moreover, we will aim to reach a consensus. If this is not possible within an agreed timeframe, we will use a majority vote for routine project decisions and record the outcome and reasons in our shared notes. Additionally, questions about assessment requirements will be referred to the tutor for clarification.

For tensions involving workload, missed commitments, or communication, the members involved will first discuss the issue privately. If necessary, another group member agreed upon by both sides will help facilitate a constructive conversation. We will agree on specific actions, responsibilities, and a follow-up date, while considering individual circumstances and maintaining a fair distribution of work. If the issue remains unresolved after these steps, repeatedly affects progress, or involves serious disrespect, we will seek support from the tutor or unit convenor. We will provide a factual summary of the issue and the steps already taken, allowing those involved to explain their perspectives.

H. Team Expectations:

- Academic Excellence & Code Standards:
    - Every member is expected to write clean, well-commented Python code, maintain reproducible Jupyter Notebooks/scripts, and adhere to proper Git branching workflows (feature branches and pull request reviews) before merging into the main branch.
    - All experiments—including hyperparameters, loss functions, and evaluation scores (such as ROC/AUC for face verification, accuracy for emotion detection, and liveness precision)—must be thoroughly logged so they can be seamlessly integrated into the group report and discussed during the Week 12 interview.
- Quality of Deliverables & Rigorous Testing:
    - Models must be fully trained using the provided datasets (Classification and Verification data) without relying entirely on un-trained pre-trained models, fulfilling all core project requirements including face verification, anti-spoofing, and real-time emotion recognition.
    - Comprehensive testing must be conducted prior to integration, including testing the anti-spoofing module against printed photos and mobile screen displays as specified in the project guidelines.
- Timely Delivery & Early Milestones:
    - Tasks must be completed ahead of internal deadlines (set several days prior to official Canvas due dates) to ensure adequate time for system integration into the webcam GUI, debugging, and final report compilation.
    - The final 2 to 3 days before project completion are strictly reserved for full system integration testing, performance optimization, and video rehearsal.
- Collaboration, Fairness, and Mutual Support:
    - Workloads must be distributed fairly across all 6 members, covering technical modules, GUI integration, documentation, and equal speaking time in the 5-10 minute demonstration video.
    - Members are expected to actively support one another, share model checkpoints, document input/output formats clearly for smooth handovers, and provide constructive, respectful feedback during code reviews and group discussions.
- Accountability & Professional Engagement:
    - Members must maintain transparent communication, respond to team inquiries within agreed timeframes, attend scheduled check-ins punctually, and update the shared task tracker regularly.
    - Everyone commits to honest peer evaluations and active individual contributions, ensuring that the final submissions including the group report, individual reports, and interview preparation—accurately reflect a solid, hands-on understanding of the implemented machine learning techniques.